

Day 4: Authentication Verification (SPF/DKIM/DMARC) & Domain Intelligence

Key Milestone: Email Authentication Verification Engine

Sprint Focus: Execution of core deliverables for Day 4 across AI/NLP, Header Forensics, IOC Extraction, Graph Correlation, Case Management, and UI Integration.

Daily Deliverables: SPF/DKIM Verification Subsystem, Typosquatting Detection Algorithm, Interactive Nodal Canvas.

Individual Team Member Task Breakdown & Contributions

Team Member & Lead Role	Domain & Ownership Area	Tasks Executed & Deliverables on Day 4
Muniswami.Y AI / NLP Lead	Explainable AI, Intent & Phishing Detection, Social Engineering signals	Added sentiment analysis and social engineering urgency cue detection (fear, scarcity, financial inducement, time pressure).
Jeevan reddy.G Header & Forensics Lead	Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis	Engineered cryptographic signature validator for DKIM (`rsa-sha256`) and TXT record parser for SPF mechanisms (`ip4`, `include`, `all`).
Lahari.K IOC & Infrastructure Lead	URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection	Implemented lookalike domain detection (typosquatting, homoglyph replacement, sub-domain tunneling) against Alexa Top 10K brands.
Hemanth.P Graph & Campaign Intelligence Lead	Attack Relationship Graph, Similarity detection, Campaign grouping & correlation	Built graph edge weighting logic: connecting disparate emails sharing identical originating IP subnets (/24 CIDR) or registrant names.
Abbas.N Risk, Case & Response Lead	Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions	Implemented threat level classification logic (`VERIFIED CLEAN`, `LOW`, `MEDIUM`, `HIGH`, `CRITICAL THREAT`) with actionable remediation steps.
Moulaanbee.N Frontend + Assistant + Simulator Lead	Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX	Implemented the interactive Attack Infrastructure Graph visualization using HTML5 Canvas and SVG nodal physics.

Integration, Verification & Quality Assessment

Cross-Module Integration:	Verified SPF/DKIM validation against real-world test emails with passing, soft-fail, and forged signatures.
Quality Assurance Status:	All code unit tests and interface contracts for Day 4 verified clean. Git commit pushed with zero blocking errors.